

GrainHero & TEQrock

AI-driven industrial IoT platform predicting and preventing post-harvest grain spoilage across Pakistan and emerging markets.

\$1.1B

ANNUAL LOSS ADDRESSED (PK)

5

GRAIN TYPES SUPPORTED

96.8%

STORAGE CAPACITY UNMONITORED

PROGRAM

NICAT Cohort 7

STATUS

Pilot Phase Active

PREPARED BY

TEQrock / GrainHero Founding Team

01 THE PROBLEM

Pakistan loses 4.5–7.5M tonnes of grain annually (15–25% of production) to post-harvest spoilage — PKR 250–350 billion (~\$1.1B) lost every year in unmonitored facilities. Modern silo storage covers only 3.2% of national capacity. Global post-harvest loss exceeds \$1.3 trillion annually (FAO, 2023).

Statistic	Value	Source
Annual wheat production	27–30M tonnes	USDA GAIN 2025
Modern silo storage coverage	~3.2% of production	PBC Research 2024
Post-harvest loss rate	15–25%	FAO FLW Database
Economic loss	PKR 250–350B (~\$1.1B)	2026 market prices
Storage gap (unmet capacity)	22M+ tonnes	PBC Research 2024

Barrier	Current Reality	GrainHero Fix
Cost	Global IoT: \$5,000–40,000/silo	\$300 hardware + \$99/mo SaaS
Power	8–14h loadshedding daily	Solar + UPS + 21-mo battery pods
Connectivity	Unreliable rural WiFi	LoRaWAN + offline SD buffer
Expertise	No technical staff on-site	Zero-config sensors, one-tap app
Language	English-only platforms	Urdu + Arabic UI

02 THE SOLUTION

GrainHero deploys wireless sensor pods inside grain stores, runs an ML ensemble for real-time spoilage-risk prediction across five grain types (rice, wheat, maize, sorghum, barley), triggers automated fan aeration, and delivers advisories via a web dashboard with insurance claim management and audit logs.

Feature	Status	Value to Customer
Real-time temp / humidity / gas monitoring	Working	Know exactly what's happening in the silo
AI spoilage risk prediction (3 classes)	Working	7–14 day early warning before visible damage
Automated fan / aeration control	Working	Physical intervention without human action
Multi-grain support (5 types)	Working	One platform for all crops
Insurance claim management	In Progress	Digital evidence trail, automated claims
OTA firmware updates	In Progress	Remote updates, no site visits
AI Research Advisor (RAG)	Planned	Gemini-powered, literature-grounded advisor

Key advantages: AI ensemble prediction (grain-specific, explainable) • Frugal hardware (<\$80/pod vs \$500+) • Hermetic bag IoT monitoring — extends shelf life 12–18 months • Offline-first (SD buffer + LoRaWAN) • Urdu + Arabic UI • Integrated insurance module.

03 MARKET ANALYSIS

Total Addressable Market

Market	Facilities	Price Point	TAM	Priority
Pakistan grain warehouses & mills	~9,200	\$99–199/mo	~\$12M/yr	Year 1
Middle East grain reserves (KSA, UAE)	~3,500	\$299/mo	\$12.6M/yr	Year 2–3
Sub-Saharan Africa (cooperatives)	~50,000+	\$29/mo	\$17.4M/yr	Year 3–5
Global infrastructure potential	—	—	\$2.1B+/yr	Long-term

Pakistan Market Segments & Competitive Landscape

Segment	Size	Willingness to Pay	Strategy
Flour mills	~2,000 (Punjab)	High	Primary pilot target
Agri cooperatives	~500 active	Medium	Group purchase discounts
PASSCO / Government	1.3M tonne cap.	Low	Year 2 formal tender
Private traders	~5,500 sites	Low	Starter tier pricing

Competitor	Geography	Price	AI?	GrainHero Advantage
Bin-Sense (Canada)	N. America	\$10K–40K/bin	No	15× cheaper, AI included
SiloBoss (Australia)	AU/NZ	\$8K–25K/silo	No	Frugal design, EM market fit
StorMax (India)	India/SE Asia	\$200–500/yr	No	ML actuation, Pakistan-specific
Grain Guard (AU)	Australia	\$5K–15K	No	6–10× cheaper, solar native
GrainHero	PK → Global	\$49–299/mo	Yes	All 6 differentiators combined

04 REVENUE MODEL & UNIT ECONOMICS

Subscription Tiers & Hardware Bundles

Tier	Price/Mo	Included	Target
Starter	\$49	2 silos, basic alerts, heuristic risk	Small farmer, trader
Professional	\$99	10 silos, AI predictions, PDF, insurance	Warehouse owner
Enterprise	\$199	Unlimited silos, SHAP explainability	Flour mills, co-ops
Platform	\$299	White-label, API access, reseller rights	Distributors

Bundle	Contents	Price
Starter Kit	4 sensor pods + LoRaWAN gateway	\$300
Professional Kit	8 pods + gateway + fan relay	\$600
Enterprise Kit	20 pods + 2 gateways + UPS	\$1,200

Unit Economics (Pro Tier)	Value	Revenue Projections	Customers	Revenue
MRR per customer	\$99	Year 1 — Pakistan	15	\$22,500
Year-1 revenue/customer	\$1,488	Year 2 — PK + UAE	70	\$196,000
Gross margin	~82%	Year 3 — PK+ME+Africa	200	\$750,000
CAC (Pakistan)	\$150–400			
LTV (3-yr, 85% renewal)	\$3,018			
LTV/CAC ratio	7–20×			
Break-even customers	19			

Additional streams: insurance referral commission (10–15%), export traceability certificates (\$0.50–1.00/tonne), solar hardware sales (35% margin), regional data licensing (Year 3).

05 PARENT COMPANY: TEQROCK

GrainHero is the flagship product of **TEQrock**, an emerging-market infrastructure holding company deploying AI/IoT solutions where traditional infrastructure fails. TEQrock’s long-term vision: a portfolio of AI/IoT products across agriculture, energy, and industrial sectors —

sharing GrainHero's hardware, cloud, and billing layer. GrainHero is the sole active product.

06 GO-TO-MARKET STRATEGY

Milestone	Timeline
Pilot silo live at partner flour mill	Month 1–2
First 3 paying customers (flour mills)	Month 3–4
15 paying customers; trade show presence	Month 5–8
Urdu/Arabic UI live; UAE first customer	Month 9
IGNITE grant application submitted	Month 9–12
40 customers; Series A preparation	Year 2 Q1
Africa pilot (Rwanda entry point)	Year 2 Q3
Series A close at \$750K ARR	Year 3

Sales approach: Direct to flour mills via live pilot-silo demos (a single 5-tonne loss = PKR 500,000, ~5 months of subscription — ROI in under 30 days). Distributor network at 15–20% channel commission. Government/PASSCO formal tender in Year 2 using pilot data as proof-of-concept.

07 RISK ANALYSIS

Risk	Mitigation
ML accuracy on real-world data lower than synthetic	"AI-assisted" marketing framing; pilot data retrains models
Cloud infrastructure cost changes	Platform built on portable, open standards — can migrate providers
Key-person dependency (small founding team)	Full technical architecture documented internally
Hardware supply chain disruption	90-day component buffer; dual-sourced MCU options

08 FINANCIAL SUMMARY & FUNDING ASK

Round	Status	Year 3 Scenario (200 customers)	Value
Bootstrapped	Active	Total ARR ~\$2,000 (hardware + cloud)	\$750,000
NICAT Cohort 7	Active	Gross margin	~82%
IGNITE Grant	Planned Q4 2026	Annual operating income target	~\$136,800
Pre-Seed / Angel	Open	Net operating income	~\$478,200
Series A	Year 3	Equity value target	~\$7.5M

Sources: FAOSTAT, USDA GAIN 2025, PBC Research 2024, FAO FLW Database. Prepared by TEQrock / GrainHero Founding Team.